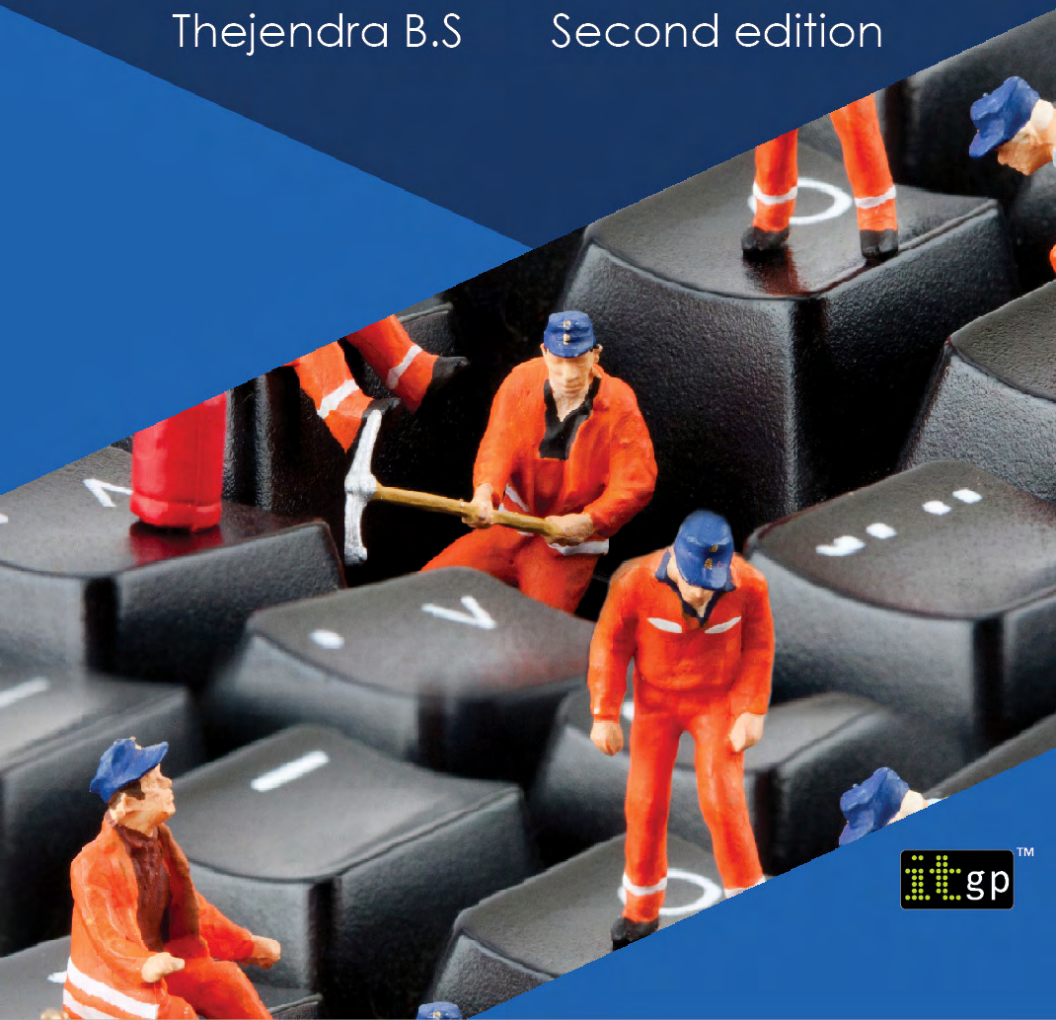


Practical IT Service Management

A concise guide for busy executives

Thejendra B.S

Second edition



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IT Governance Publishing

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All ITIL definitions in this book are taken from the official ITIL® Glossary. The full glossary is available online - links are provided in the appendix.

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PREFACE

Practical IT Service Management is a concise guide to implementing a professional, technical service management structure in your organisation, based on the international best practice framework ITIL[®] (Information Technology Infrastructure Library[®]). This framework is globally the most widely accepted approach to technical service management, and is developed based on input from several public and private sector organisations. This book explains the fundamentals of the latest ITIL 2011 version and its implementation in an easy, self-study approach for all technical and business staff in your organisation. The entire book is written in a question and answer format for easy comprehension and speedy reading. Each chapter covers just one specific area of ITIL, and each topic is explained concisely, with very few answers extending beyond one page. Practical and real-life examples are used throughout. *Practical IT Service Management* is designed to be a stepping stone to the official books on ITIL published by the The Stationery Office (TSO).

Unless stated otherwise, the names of the companies and people mentioned in the examples in this book are fictitious. The names of actual companies and products mentioned are the trademarks of their respective organisations. I would like to thank Alan Calder and Vicki Utting for their immense help in preparing this book. Although this manuscript has been prepared with the utmost care, the author, publisher, editor, or any other party associated with this book, can accept no liability for any direct or indirect damages caused by following the advice given here. However, suggestions

Preface

for improvement, errors or mistakes observed, corrections required and any other relevant information that could be incorporated in a future edition, will be gratefully received at thejendra@yahoo.com or thejendrabs@gmail.com.

Thejendra B.S

January 2014

ABOUT THE AUTHOR

Thejendra B.S is an IT Manager for a software development company in Bangalore, India. Starting work as a field engineer, he has more than 20 years' experience in a wide range of roles in IT areas. He has been involved in IT support, help desk, DRP-BCP, asset management and IT security, and has implemented numerous small to large IT projects worth millions of pounds. Thejendra has worked in India, Saudi Arabia, Dubai, Bahrain, Qatar and Australia. During his years in the IT world, he has dealt with countless flavours of customers, vendors, end-users and organisations of all sizes. His articles have also been published on reputed websites, such as geekleaders.com, drj.com, ezinearticles.com, cio.com, techrepublic.com, cnbc.com, sourcingmag.com and many ezines.

Thejendra can be contacted at thejendra@yahoo.com or thejendrabs@gmail.com. For further details of his books and articles, please visit his website www.thejendra.com.

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INTRODUCTION

‘A business absolutely devoted to service will have only one worry about profits. They will be embarrassingly large.’

Henry Ford

What is this book about?

The advancement and ease of availability of new and useful technologies has enabled thousands of organisations, worldwide, to implement, and become heavily dependent on, technology for running their businesses. It is not possible to run any organisation, small or big, without the use of some computer or telecom-related technologies. With so much proliferation of hardware, software and networking equipment it is necessary to have specialised and dedicated technology support departments to look after them. Otherwise, companies can get into serious trouble. A professional, technology support department is as essential to any organisation as a qualified finance department or a senior management team. Although organisations are free to have their own proprietary flavours of technical support, suiting their needs, it is always better to adopt some international best practices, as they prevent organisations from reinventing the wheel.

This concise book explains how to implement one such international best practice, ITIL. ITIL is a Registered Trademark, and all its contents are owned by the Cabinet Office, under the HM Government. Earlier ownership was with OGC (Office of Government Commerce). The

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publications (books) continue to be Crown Copyright. In addition, the Cabinet Office has entered into a joint venture with Capita plc. to commercialise its accreditation and publishing services, including ITIL. The new name for this joint venture company is called AXELOS, and it will inherit the entire portfolio of best management practice products from the Cabinet Office. The joint venture is expected to become fully operational by the middle of 2014. The term Axelos comes from a Greek-French philosopher named Kostas Axelos, who united old and new with games and openness.

All the ITIL concepts can be freely adopted by anyone implementing IT Service Management (ITSM) within their organisation. Written in a condensed style, *Practical IT Service Management* explains how you can interpret and implement the ITIL concepts in an easy, self-help format.

Who should read this book?

This book is written for busy IT executives in any organisation. Most busy executives often don't have the time, patience, luxury or interest to read detailed, academically-oriented IT books, due to their never-ending workloads and competitive pressures. They need quick, practical information on a topic or concept that will help them in their workplace. This book fills the needs of such an audience and will be of use to:

- Technical managers
- Technical support specialists
- IT consultants
- Chief technical officers

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- Chief information officers
- Business managers
- And even members of the Board of Directors.

CHAPTER 1: INTRODUCTION TO IT SERVICE MANAGEMENT

What is IT?

The term 'IT' is an abbreviation of Information Technology. A general dictionary defines IT as the development, installation and implementation of computer systems, telecommunications and software applications. In practical terms, IT consists of:

1. Computers, such as desktops, servers, laptops, mainframes, and the data that they hold.
2. Software, such as operating systems (Windows, Unix, Linux, Novell, specialised operating systems) and applications, such as word processors, spreadsheets, databases, productivity tools, business applications and custom-built applications.
3. Communication and telecom equipment, such as PBX, lease lines, the Internet, telephone networks, Local Area and Wide Area Networks.
4. Other specialised IT equipment and software.

The ITIL definition of IT is:

'The use of technology for the storage, communication or processing of information. The technology typically includes computers, telecommunications, applications and other software. The information may include business data, voice, images, video, etc. Information technology is often used to support business processes through IT services.'

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What are IT services?

The term ‘IT services’ refers to a set of support and maintenance functions provided by technically qualified staff (internal or outsourced) to an organisation that uses various computers, software, printers, hardware and communication facilities. An IT service may range from providing access to a simple application, such as a word processor for all end-users, or access into a complex network consisting of hundreds of different types of computers, operating systems, servers, e-mail systems, websites, databases, telecom systems and Internet access used by hundreds of end-users inside an organisation.

The ITIL definition of an IT service is:

‘A service provided by an IT service provider. An IT service is made up of a combination of information technology, people and processes. A customer-facing IT service directly supports the business processes of one or more customers and its service level targets should be defined in a service level agreement. Other IT services, called supporting services, are not directly used by the business but are required by the service provider to deliver customer-facing services.’

What is IT service management?

The term *IT Service Management* refers to an orderly and professional method followed by an IT department to provide reliable and efficient information systems and support to meet your business requirements. Most organisations now understand the benefits of having IT throughout their internal environment, but do not understand the need for managing it properly. If IT equipment and services are not managed correctly in your organisation, you could get into serious trouble. Firstly, and as mentioned

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earlier, no modern organisation can run its operations, or survive, without using one or more computers, software, telecommunications and the Internet. If an important computer system stops working, then business may have to stop if it is not possible to switch over to alternative manual processes for any length of time. Secondly, computer systems and networks are extremely complex and complicated for any business person to maintain, or support, on their own. Specialised employees are required who understand how those systems work and how to babysit them. IT services should be in alignment with your business strategy and objectives. From a simple nuts and bolts perspective, IT service management means that the ‘techies’ (employed or outsourced) in the organisation are professionally managing and maintaining the computers, networks, telecommunications, data storage and retrieval, e-mail systems and databases, owned or used by your business.

The ITIL definition of IT service management is:

‘The implementation and management of quality IT services that meet the needs of the business. IT service management is performed by IT service providers through an appropriate mix of people, process and information technology.’

What problematic issues do IT departments commonly face?

Running an IT department is a herculean task. There will always be difficulties and headaches to keep the staff perpetually busy and hassled. Some of the common obstacles faced by IT departments of many small, and even large, organisations include the following:

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- Roles and responsibilities of staff are not clearly defined or are non-existent. No structured customer support mechanism is in place. No help desk or service desk facilities.
- Business managers do not understand (or try to understand) the technical department's work and constraints, and technical people do not understand (or try to understand) business people's needs.
- A single IT person or an IT team that is too small are responsible for anything and everything related to IT. Excessive workloads and poor career growth prospects.
- Lack of clearly defined and simple processes. No service level agreements, vendor agreements and technical training.
- Frequent disagreements between business and IT departments for service and cost expectations.
- Business and technical staff not seeing eye to eye. Poor management buy-in, inadequate funding, culture issues and resistance to change.
- Businesses not understanding the essential requirements for using IT in their organisations (proper IT staffing, exponential hardware and software budgets, ongoing costs and frequent and necessary upgrades).
- Technical staff concentrating only on technical matters, and unable, or unwilling, to understand business needs.
- No proactive IT problem prevention methods. Only reactive support. Issues get solved after they occur, with no prevention mechanism in place.

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- IT staff using outdated tools and equipment, resulting in the IT department being out of sync with modern business demands.

What issues do businesses face through heavy dependence on IT?

In the 1990s only very large organisations could afford to use computers. At that time, IT was not considered as being essential to run a business. This thinking is no longer applicable, and IT has proved its benefits, even in the smallest of organisations. However, using information technology is a catch-22 situation, as businesses have become excessively dependent on IT. You cannot live with IT, nor can you live without IT. With so much dependence on IT there will be associated risks and issues. To fully answer this question, it is first necessary to understand how IT normally gets implemented in an organisation. Many organisations can easily buy the necessary computers, software and telecommunications for running their businesses. However, the implementation of IT is often carried out without proper planning of any sort due to numerous reasons, such as a lack of appropriate knowledge.

There will be several IT-related issues that will cause minor to major irritations, or even bring an organisation to an abrupt halt. The following examples show how many organisations implement IT and the hair-raising issues they can face.

Example of poor IT implementation:

The owner of a small business may buy a single computer, initially for general use. After discovering the benefits of using a

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computer, he may immediately decide to buy 25 more for his staff.

Within a short time his business will be computerised, and very soon IT support headaches will enter the business. Using a computer may be easy, but maintaining a computer system is a complicated task. Users may suddenly experience crippling virus attacks, equipment failures, software licensing issues, data corruption, data loss, back-up issues and upgrade issues. They may not be in a position to support and maintain a computer network and its associated functions. Overnight, a smart purchasing assistant may undergo a crash course in computer maintenance, or buy a book on *Computer Maintenance*, and soon will be given responsibility for the technical support of the business, along with his or her other responsibilities. IT departments begin their life in this way in many organisations. However, this sort of approach will lead to major and uncontrollable issues later on.

Example of poor and inadequate IT support:

Let us take a simple example of how a single hard disk crash can cripple your organisation. A technician who lacks business sensitivity may view a computer hard disk crash as a simple issue, whereas it may be seen as a critical issue for the business owner since the entire business and financial data may be on the failed disk. To add to the misery, the disk was probably not being backed up regularly. On the other side of the coin, the business owner may have earlier refused to invest money on an essential device, such as a tape drive for data back-up. Both parties will blame each other. A typical interaction between the help desk and the business departments in many organisations can be like this:

Finance Department: ‘Hello. Our finance server is not working. Can you fix it?’

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Help desk: ‘Which one?’

Finance Department: ‘The one that we use in our department. It’s a black system with a green keyboard.’

Help desk: ‘I had a look at it, but the hard disk is dead and we will have to replace it. I will call the vendor and arrange for a replacement if possible.’

Finance Department: ‘What about our data?’

Help desk: ‘I’m afraid we can’t recover the data. The disk is dead and we have not been backing up the data of that server, because nobody told us to. Finance did not approve the purchase of a tape drive for this machine.’

Finance Department: ‘Oh no. We have our entire payroll, purchasing, billing, sales and other important financial data for the entire company on that machine. Five years of data!’

Help desk: ‘Unfortunately there is nothing we can do. Please excuse me, I have to go and attend another call.’

A situation like that can cripple your organisation within hours.

Other common IT headaches

Other IT-related frequent pin pricks and shocks can be as follows:

- Your end-users don’t know who to contact when their computers and other IT equipment fail.
- Your techies attend end-user calls if they can, when they can.
- Business managers do not understand why their IT infrastructure is always having disruptions of a similar nature.
- Monday morning chaos. All computer systems are down

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for some reason. The IT department probably performed some maintenance activities over the weekend.

- Businesses cannot commit about their products and services to their external customers. (*See next example.*)
- Your end-users do not know if all IT services will be available for them every day to complete their activities.
- Viruses, crippling and lengthy IT shutdowns, are common.
- End-users always face a shortage of computers, disk space, data corruption and data loss.
- Business managers do not know why they need to shell out another bag of cash for some software the IT department needs.

Example of IT breakdown affecting business:

New Sales Manager: ‘Folks, where are you going? That sales quote must be sent to the Abacus Company today or we lose that account.’

Sales Team: ‘We are going home. All computers are down. We can’t prepare the sales quote without a computer.’

New Sales Manager: ‘Home? When will the IT department fix it?’

Sales Team: ‘They said it may take a couple of days or more.’

New Sales Manager: ‘What? We can’t wait that long.’

Sales Team: ‘We do here. Very often it can take three days to fix IT issues. Computer breakdowns are quite frequent here.’

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New Sales Manager: ‘This isn’t good enough. I have committed the pricing quote to the Abacus Company. If I don’t send it today we may lose a £50,000 order.’

How can professional IT service management help?

The above examples show what could happen to your organisation without a professional and proactive IT service department. Without such a department, your organisation could face crippling and profit-threatening situations. However, if you have implemented professional IT service management practices, such harrowing situations are less likely to occur. Even if an important disk has crashed, with professional IT service management it should be able to be restored in a matter of hours.

Haphazard IT support directly, or indirectly, impacts your main business. For example, will external customers open an account in a bank that has frequent and lengthy computer breakdowns, virus attacks and shutdowns? Or if your entire manufacturing operation is computerised, and if there are frequent IT breakdowns, think about the losses, delay and its business impact. Considering the complexity and importance of today’s computer systems, it is imperative to bring in some measurable and verifiable IT service standards so that your business managers understand, amongst others, the IT department’s scope of work, the deliverables, the constraints, the limitations and budgetary needs. It is absolutely vital for both IT and business departments to understand that the quality of support, its availability, and the recoverability of your IT infrastructure, will directly influence the quality, profitability and respectability of your organisation.

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This is where professional IT service management, or ITIL, can help because these are industry best practices that can safeguard your organisation. Otherwise, your organisation may follow some proprietary service processes and methodologies that may, or may not, save you and also not be transparent, documented, controlled, measurable, repeatable and portable. By implementing IT service management, your business owners can have the satisfaction and peace of mind of knowing that the IT infrastructure necessary for running your business is in safe hands.

However, implementing a best practice IT service management system does not mean discarding every current process or method used by your organisation and starting from scratch. You can gradually implement an IT service management system that will be much more efficient than a method developed in-house.

CHAPTER 2: OVERVIEW OF ITIL 2011

‘Call them rules or call them limits, good ones, I believe, have this in common: They serve reasonable purposes; they are practical and within a child’s capability; they are consistent; and they are an expression of loving concern.’

Fred Rogers

This chapter provides an overview of ITIL 2011, which is the latest update released. Earlier versions will also be explained briefly, but elaborate comparisons will not be made between them, to avoid confusion. As mentioned earlier, ITIL is under copyright. No exact material, diagrams or contents from the official ITIL books are reproduced in this book, however, the exact ITIL definitions from the official ITIL glossary are mentioned in some chapters, just to elaborate the concepts being explained.

What is ITIL?

The term ‘ITIL’ refers to a best practice framework for IT service management, and consists of a series of official publications (books) giving exhaustive guidance on how to provide quality IT services in your organisation. The books explain the various processes and kinds of departments needed to support these IT services. As mentioned before, the best practice guidance contained in ITIL can be freely used by any organisation of any size, operating in any industry.

However, as soon as the word ITIL is mentioned, many overloaded and overburdened IT departments and business

2: Overview of ITIL 2011

people start imagining all sorts of scary views, such as it is a bureaucratic process, it is very complex, it must be highly theoretical, which can only add to their woes, rather than reduce them. For IT managers involved in practical, technical aspects, the thick official ITIL books may seem like a detailed collection of dull processes, raising doubts on how they can actually use them in the real world. Also, many business people, IT departments and managements of small and medium organisations live in the misconception that ITIL is beyond their expertise or affordability, and perhaps applicable only to large organisations.

All the above fears are understandable because ITIL is created and maintained by a government organisation, unlike Six Sigma which was created by a Fortune 500 organisation, led by a glamorous CEO. After all, governments worldwide are known for their lethargic, obscure and bureaucratic processes that only delay and make things complex. However, as you can never judge a book by its cover, you should not judge ITIL by who created it. ITIL is not the scary stuff that most businesses imagine. Actually, ITIL is a lot of practical IT management common sense and not just some impractical theories. Generous doses of ITIL can be implemented by practically any organisation (small, medium or large) to bring some law and order to their IT infrastructure management. By implementing ITIL, you can avoid, or eliminate, various IT issues, and bring a very high degree of stability and predictability to your IT infrastructure. ITIL offers value and return on investment to every business owner, service provider, CIOs, CTOs and CEOs. However, extracting the wealth of advice given in ITIL actually depends on how you can *interpret* the given processes for practical purposes and apply them to your

2: Overview of ITIL 2011

organisation. This book aims to help you achieve that *interpretation* in a short and realistic way.

Before you begin the actual journey of IT service management based on ITIL, it is necessary to understand the meaning of certain common words, such as ‘**business**’, ‘**customer**’ and ‘**end-user**’, as used in the IT service management world. These are all core concepts of ITIL and hold good for all ITIL versions. This can be explained through an example.

Example: The RockSolid Corp:

Assume there is a modern company called RockSolid Corp, owned by Mr Johns, that manufactures and sells industrial air conditioners (a/c). The company has about a thousand employees spread among several departments, such as sales, finance, human resources, a/c engineering, clerical support, a/c technical design, support and a/c servicing. All these employees have been provided with a desktop computer, access to e-mail, telephones, the Internet, office applications and other software necessary for their departments. All are connected by Local and Wide Area Networks. There is also a separate department called the Technical Services Division, consisting of several employees (and also outsourced staff, such as technical contractors and consultants) trained and responsible for maintaining those computers, e-mail systems, Internet access software and networks. The Technical Services Division, in turn, has several sub-departments, or groups, specialising in a particular IT area, such as desktop support, network support, software support, service desk, data centre, IT strategy, planning and operations. All the departments of the Technical Services Division are located within the buildings of RockSolid Corp.

In addition, this Technical Services Division and its departments also have support and consultancy arrangements with various

2: Overview of ITIL 2011

external vendors and specialists, ranging from common hardware and software suppliers, to ultra-specialist companies that can implement and maintain massive technical projects for the organisation. Such arrangements will be required because companies can no longer afford to employ all types of technical experts on their payroll. For example, RockSolid Corp may have a multimillion pound long-term contract with a massive telecommunications company (say ABC-Telecom) to meet and maintain all its telecom and networking needs on an end to end basis. In such cases, ABC-Telecom will be actively involved in some RockSolid Corp activities, such as decision making on technical upgrades, decommissioning obsolete equipment, providing support, maintaining infrastructure inventories, running telecom projects and providing account managers to route requests and generate reports. In such cases, ABC-Telecom can be called a 'technology partner' or an 'IT Service Provider' who will be dedicating a set of their employees to service RockSolid Corp.

As you may have observed, most large companies enter into long-term contracts with several IT service providers, to minimise having in-house expertise. A large bank or supermarket chain need not employ all types of technical specialists. Instead, they can hire, or enter into a long-term contract with, major IT service providers to meet all their needs.

According to ITIL, the following words have the following meanings.

- **Organisation:** means the RockSolid Corp, with all its employees, equipment, etc.
- **Business:** The primary business of RockSolid Corp is selling industrial air conditioners. Without selling air conditioners, the organisation cannot exist. As a rule, every air conditioner purchase order is processed and

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shipped within two business days.

- **IT services:** covers all the IT facilities and access to various applications, computers, telecommunications, software, databases, e-mail systems, web servers, associated infrastructure, and the Technical Services Division (with all its sub-departments) who look after that equipment in the RockSolid organisation. All the departments of the Technical Services Division are collectively, and collaboratively, responsible for maintaining the IT infrastructure of RockSolid Corp.
- **Customer:** This term has to be used carefully. A customer in ITIL is the person (or senior management) who pays for, and owns, the IT Services or the Technical Services Division. Typically, this is someone who is responsible for paying, or absorbing the cost of having an IT service within his or her organisation. In this case, the owner of RockSolid Corp, Mr Johns and his business managers, are the customers for IT services because they are paying money to own, or hire, the Technical Services Divisions. In turn, Mr Johns will have external customers who have purchased air conditioners manufactured by RockSolid Corp.
- **End-users:** The various employees and departments, such as sales, finance and HR within the RockSolid organisation who use the IT services on a day-to-day basis, are called end-users. They depend on IT services for running the business smoothly. The finance department will depend on IT services to maintain and support their payroll and accounts server. The engineering department will depend on IT services to maintain and support their design server. In many companies, end-user departments also pay, or get

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charged, for having regular or special services from IT services. End-users can also be treated as a flavour of customers of IT services from a broad perspective.

To summarise:

- Customers for IT services of RockSolid Corp are the business managers (who pay for having the IT services) and end-users who may directly, or indirectly, pay for these services. This book will deal mainly with these customers and look at how IT services provide effective internal support to the end-users, business managers and divisions inside the organisation(s). Whenever and wherever we mention the word customer in this book, it will always mean the employees and management of RockSolid Corp.
- Customers for RockSolid Corp are the various external companies and individuals who have purchased air conditioners. In this book we will not refer to these external customers.

What are the main benefits of using a framework, such as ITIL?

Many organisations believe they have already implemented excellent self-developed IT services and don't need to change, as the current framework might be acceptable to the business imperatives. However, on closer examination, these customised services will usually lack many necessary and essential processes that could enhance the IT department. The benefits of using professional IT service management processes, such as those in ITIL, are simply enormous:

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- ITIL is a framework that offers benefits that demonstrate value and return on investment to every business owner, service provider, CIOs, CTOs and a CEO.
- Proven and tested processes. No need for businesses to reinvent the wheel for implementing IT services in their organisations. Covers end-to-end.
- Non-proprietary practices. Though ITIL is owned by HM Government, it does not require a license to practice, and it is independent of any commercial solution or platform. Every organisation can use the official ITIL books to implement the processes.
- Improved quality of IT service for business functions. Reduced downtime, improved customer and end-user satisfaction.
- Measurable, controllable, recoverable.
- ITIL is scalable. It can be adapted for any size of organisation.
- Proactive rather than reactive. Clearly defined roles, responsibilities and activities.
- Greater understanding of IT and its limitations. Business will understand IT better and vice versa.
- There are a range of accredited ITIL training and education courses. This has resulted in the growth of a number of support services, training institutes, tools and consultancy services that can help your organisation's IT departments.
- Return on Investment (ROI). ITIL helps IT departments demonstrate their return on investment and measurable value to the business, and also cut IT costs. This helps

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establish a business case for new, or continuing, investment in IT.

- ITIL also helps in outsourcing. ITIL is widely practiced among many industry service providers and they can easily help your organisation's IT departments.
- Continuous improvement, stability and proactive problem prevention.
- Improved business image. Businesses will also learn what to commit, and what not to commit, to their external customers.

Important note

While ITIL has many benefits, it is not prescriptive. The important point is that the framework is independent of any IT vendors or their proprietary systems. As it is vendor neutral, it does not recommend or criticise any vendor's products or practices. This means ITIL will not tell you to use Microsoft® Windows® or buy Cisco® routers. Organisations are free to choose whatever IT equipment is best suited to their businesses. ITIL does not lay down any rigid guidelines or strict rules, but focuses on IT service management best practices that can be interpreted and used in various ways, or customised to your needs. You should understand what works in one company's IT environment may not work exactly the same in another. But even if your organisation implements only part of the ITIL framework, you can put your organisation on the road to becoming world class.

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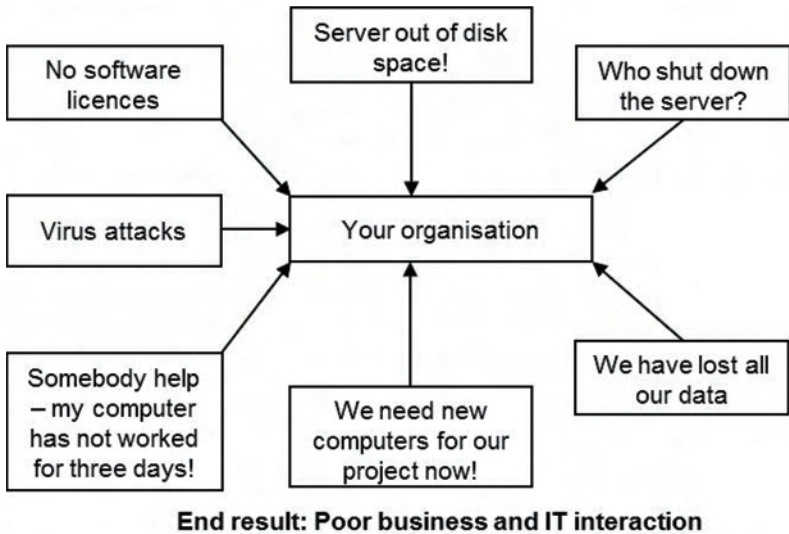


Figure 1: Before IT service management implementation

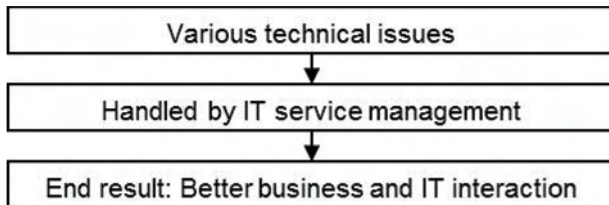


Figure 2: After IT service management implementation

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How many departments are needed to implement IT service management or ITIL?

Professional IT service management, in any organisation, will involve several processes and functions over a period of time. Ideally, it is best to have separate departments to handle each process and function. However, it may not be possible, or affordable, for all organisations to build so many additional departments and personnel to handle all IT areas. If an organisation is big and has thousands of employees, then it is strongly recommended that it has separate departments with the workforce and budgets to handle their respective processes. If an organisation is small, then it will need to investigate whether it actually needs all the processes in the first place. A smaller organisation can choose to implement only a subset of all the processes, perhaps only the basic processes to begin with, and to have a diluted version of the others until it is necessary or affordable to implement them fully. Alternatively, it could have a single department with several managers who can become individual owners of each of the processes.

For example, IT services in RockSolid Corp (*see aforementioned example*) will consist of separate departments handling each of the processes. As a rule of thumb, small organisations can implement a workable ITIL set-up within three to nine months. Bigger organisations may take more than a year to implement essential ITIL. Fundamentally it depends on your internal speed, your willingness, and other political considerations you may need to overcome to truly implement such a framework.

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How did ITIL start and evolve?

History of ITIL

During the late 1980s, the Central Computer and Telecommunications Agency (CCTA) started to work on a set of processes called the Information Technology Infrastructure Library (ITIL). Soon, many private and public companies adopted the framework and added their own best practices, and thus the ITIL framework became a set of industry best practices and a de facto standard in delivering IT services for all types of organisations.

In 2000, Microsoft® started using ITIL as the basis of the Microsoft® Operations Framework® (MOF) to support the launch of their Datacentre product.

In 2001, ITIL Version 2 was released with the Service Support and Service Delivery books.

In 2007, ITIL Version 3 was released (replacing Version 2) but it still included all the core principles and fundamentals of Version 2. Version 3 was organised around the concept of the service lifecycle, and included several new benefits to help businesses become world class.

By 2010, Version 2 and its certifications were fully withdrawn and Version 3 underwent various enhancements to meet industry requirements. It is now called ITIL 2011. However, ITIL officials have clarified that the ITIL 2011 edition is not a completely new version. In fact, it is only an update, but substantial content has been added to make ITIL easier to teach.

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What were the main processes of ITIL Version 2?

ITIL Version 2 consisted of the following processes, each of which had many sub-processes:

- Service support
- Service delivery
- Planning to implement service management
- Applications management
- ICT infrastructure management
- The business perspective
- Security management.

Although Version 2 and its exams have been officially discontinued, this does not mean its concepts are outdated or useless.

What was ITIL Version 3?

Version 3 was an enhanced and improved version of Version 2 best practices. The structure and content of Version 3 was based on extensive public consultations and contributions from industry leaders, customers, users, vendors, service providers and other best practice organisations, to determine what improvements would make it suitable to modern, complex business requirements. However, this does not mean the concepts of ITIL Version 3 are outdated or useless because Version 3 was built on top of Version 2.

Version 3 had the following processes, each of which had many sub-processes and functions:

- Service strategy

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- Service design
- Service transition
- Service operation
- Continual service improvement.

What is ITIL 2011?

ITIL 2011 is a substantial update to Version 3. This update was released to resolve certain inconsistencies in the documentation, fix some errors and address suggestions that were submitted by the training community. This book will essentially cover the processes of ITIL 2011 in more detail.

In summary, ITIL Version 2 has been phased out. The term ITIL 2007 is used for the first edition of ITIL Version 3. The latest edition is now referred to as ITIL 2011, or simply ITIL.

Table 1 briefly outlines the major difference between the various versions of ITIL

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Table 1: Major differences between the various versions of ITIL

Version	Focus
2	Process based and operationally focused.
3	Value-based and business-focused service practice. Focuses on the alignment of IT and the business. Guides IT service management and organisations from just providing a great service, to becoming innovative and best in its class. A set of specialised organisational capabilities for providing value to customers in the form of services.
2011	ITIL 2011 is not a new version. It is only an update. This update was released to resolve certain inconsistencies in the documentation, fix some errors and address suggestions that were submitted by the training community for making ITIL easier to teach.

Will the books and training change with ITIL 2011?

Yes. New books have been released for ITIL 2011 editions. The official books can be purchased from IT Governance: www.itgovernance.co.uk/ITIL-books.aspx.

The new training courses and certification details can be obtained from the same website at: www.itgovernance.co.uk/itsm_learning.aspx.

Will the earlier ITIL concepts become invalid?

Not exactly, since the core principles won't change. However, you will not be able to take its exams anymore - ITIL 2011 has enhanced exams and certifications.

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What are the latest ITIL exams or certifications today?

The following certifications are now available:

1. ITIL Foundation Certificate
2. ITIL Intermediate Certificate
3. ITIL Expert Certificate
4. ITIL Master Certificate.

Each of the above certifications are briefly explained below.

ITIL Foundation Certificate: This certificate is the introductory qualification for IT service management. The syllabus covers the terminology, processes and concepts of ITIL to provide a good grounding in ITIL principles. The ITIL Foundation exam is a 60 minute exam with 40 multiple choice questions. To pass you must score 65% or more (26/40). It can be taken online, or through an examination centre, or as part of a training course. This certification provides two credits toward your ITIL Expert certification.

ITIL Intermediate Certificate: This certificate evaluates an individual's ability to assess and apply the concepts of ITIL. The qualifications are organised around two streams – lifecycle and capability. The lifecycle stream is organised around the five core ITIL publications in the ITIL Lifecycle Publication Suite. By completing a lifecycle stream course you gain three credits toward the ITIL Expert certification. The capability stream is built around four practitioner-based clusters. Upon completing the relevant capability stream course, you will receive four credits toward your ITIL Expert certification.

ITIL Expert Certificate: The ITIL Expert certificate is automatically awarded to those who have achieved a total of 22 credits. Gaining the ITIL Expert certification demonstrates an individual's superior knowledge of the ITIL

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service lifecycle in its entirety. The 22 credits have to include credits from the mandatory Foundation and Managing Across the Lifecycle units, the other 15 credits can come from a variety of other Intermediate units prior to studying for the Managing Across the Lifecycle exam.

ITIL Master Certificate: The ITIL Master qualification is aimed at service management practitioners and consultants who are highly experienced at applying, managing and improving IT service management. To be eligible for the Master certificate, you must already possess an Expert certificate and have worked in IT service management for at least five years in leadership, management or higher management advisory level. The Master certificate will require candidates to explain and justify how they selected and applied a range of principles, knowledge, methods and techniques from ITIL, and other supporting management techniques, to accomplish business goals in multiple practical assignments.

What is meant by service in ITIL?

A service is a means of delivering value to customers by facilitating outcomes that customers want to achieve. The term ‘service’ is also used as a synonym for core service, IT service or service package.

What is the ITIL 2011 library?

The ITIL library is made up of five core books which are collectively known as the ITIL Lifecycle Publication Suite. The five core books are listed below.

1. Service Strategy
2. Service Design
3. Service Transition

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4. Service Operation

5. Continual Service Improvement.

Each of these core books has multiple first level sub-processes, as outlined in the table below. There are also second level processes for many of the first level processes.

Table 2: Summary of ITIL 2011 Processes

CORE BOOK	SUB-PROCESSES
Service Strategy	1. Strategy management for IT services 2. Service portfolio management 3. Demand management 4. Financial management for IT services 5. Business relationship management
Service Design	1. Design coordination 2. Service catalogue management 3. Service level management 4. Risk management 5. Capacity management 6. Availability management 7. IT service continuity management 8. Information security management 9. Compliance management 10. Architecture management 11. Supplier management
Service Transition	1. Change management 2. Change evaluation 3. Project management 4. Application development 5. Release and deployment management 6. Service validation and testing 7. Service asset and configuration management 8. Knowledge management

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CORE BOOK	SUB-PROCESSES
Service Operation	1. Event management 2. Incident management 3. Request fulfillment 4. Access management 5. Problem management 6. IT operations control 7. Facilities management 8. Application management 9. Technical management
Continual Service Improvement	1. Service review 2. Process evaluation 3. Definition of CSI initiatives 4. Monitoring of CSI initiatives

The first level sub-processes of each of the core books will be explained in the next chapter.